

Week 6 – Service Management & System Monitoring

The purpose of week 6 is to demonstrate effective service management, system monitoring, and logging on a Linux system. This phase ensures that essential services are running securely, unnecessary services are disabled, and system activity can be monitored and audited. These tasks improve system reliability, availability, and security.

In this phase, the status of key system services was checked using systemctl. The SSH service was verified to ensure it is running and enabled at boot, allowing secure remote administration of the system:

```
studentuser@katana-VirtualBox:~$ sudo systemctl status ssh
[sudo] password for studentuser:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-12-13 13:32:24 GMT; 1h 7min ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 7691 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 7694 (sshd)
     Tasks: 1 (limit: 2255)
    Memory: 1.5M (peak: 1.8M)
       CPU: 23ms
    CGroup: /system.slice/ssh.service
            └─7694 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 13 13:32:24 katana-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Dec 13 13:32:24 katana-VirtualBox sshd[7694]: Server listening on 0.0.0.0 port 22.
Dec 13 13:32:24 katana-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Dec 13 13:32:24 katana-VirtualBox sshd[7694]: Server listening on :: port 22.
studentuser@katana-VirtualBox:~$
```

This ensures persistent remote access to the server.

To demonstrate service control, a non-essential service was identified and disabled to reduce the system's attack surface:

```
studentuser@katana-VirtualBox:~$ sudo systemctl disable bluetooth
Synchronizing state of bluetooth.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable bluetooth
Removed "/etc/systemd/system/bluetooth.target.wants/bluetooth.service".
Removed "/etc/systemd/system/dbus-org.bluez.service".
studentuser@katana-VirtualBox:~$ sudo systemctl stop bluetooth
studentuser@katana-VirtualBox:~$
```

Disabling unused services improves security by limiting unnecessary background processes.

System resource usage was monitored using real-time monitoring tools. This allows administrators to identify performance issues such as high CPU or memory usage:

```
top - 14:58:38 up 5:51, 2 users, load average: 0.00, 0.02, 0.01
Tasks: 228 total, 1 running, 228 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.9 us, 0.5 sy, 0.0 ni, 98.2 id, 0.0 wa, 0.0 hi, 0.4 si, 0.0 st
MiB Mem : 1968.3 total, 335.9 free, 1125.8 used, 669.1 buff/cache
MiB Swap: 2688.0 total, 2636.5 free, 43.5 used, 842.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S  %CPU  %MEM     TIME+ COMMAND
 143 katana    20   0 50400 49368 28172 S 11.3 2.5 1:18.55 kintreport-tray
 882 root      20   0 324124 83960 50712 S 1.0 4.2 0:53.58 Xorg
4689 katana   20   0 472888 41652 29752 S 0.7 2.1 0:06.71 xfce4-terminal
1389 katana   20   0 463576 38932 28024 S 0.3 1.9 0:01.65 xfce4-panel
8896 root      20   0 0 0 0 S 0.1 0.3 0:00.16 kworker/1:2-events
 1 root      20   0 22748 13828 9348 S 0.0 0.7 0:05.78 systemd
 2 root      20   0 0 0 0 S 0.0 0.0 0:00.02 kthreadd
 3 root      20   0 0 0 0 S 0.5 0.0 0:00.00 pool_workqueue_release
 4 root      20  -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_gp
 5 root      20  -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-sync_wq
 6 root      20  -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-kfree_rcu_reclaim
 7 root      20  -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-slab_flushwq
 8 root      20  -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-netfs
12 root      20   0 0 0 0 I 0.0 0.0 0:00.00 kworker/0:0-ip6_addrconf
13 root      20  -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-mm_percpu_wq
14 root      20   0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_kthreadd
15 root      20   0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_rude_kthread
16 root      20   0 0 0 0 I 0.0 0.0 0:00.00 rcu_tasks_trace_kthread
17 root      20   0 0 0 0 S 0.0 0.0 0:00.71 ksoftirqd/0
18 root      20   0 0 0 0 I 0.0 0.0 0:02.20 rcu_preempt
19 root      20   0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_par_gp_kthread_worker/0
20 root      20   0 0 0 0 S 0.0 0.0 0:00.04 rcu_exp_gp_kthread_worker
21 root      20   0 0 0 0 S 0.0 0.0 0:00.05 migration/0
22 root     -51   0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/0
23 root      20   0 0 0 0 S 0.0 0.0 0:00.00 cpup/p/0
24 root      20   0 0 0 0 S 0.0 0.0 0:00.00 cpup/p/1
25 root     -51   0 0 0 0 S 0.0 0.0 0:00.00 idle_inject/1
26 root      20   0 0 0 0 S 0.0 0.0 0:00.00 migration/1
27 root      20   0 0 0 0 S 0.0 0.0 0:02.08 ksoftirqd/1
32 root      20   0 0 0 0 S 0.0 0.0 0:00.00 kdevtmpfs
33 root      20  -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-inet_frag_wq
34 root      20   0 0 0 0 S 0.0 0.0 0:00.00 kauditd
35 root      20   0 0 0 0 S 0.0 0.0 0:00.02 khungtaskd
37 root      20   0 0 0 0 S 0.0 0.0 0:00.00 oom_reaper
```

This information is critical for diagnosing performance issues and ensuring system stability.

Disk usage was checked to ensure sufficient storage availability and prevent system failures caused by full disks:

```
studentuser@katana-VirtualBox:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            197M   1.4M  196M   1% /run
/dev/sda3        24G   12G   11G  53% /
tmpfs            985M    0  985M   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
/dev/sda2        512M   6.2M  506M   2% /boot/efi
tmpfs            197M  140K   197M   1% /run/user/1000
tmpfs            197M  148K   197M   1% /run/user/1001
studentuser@katana-VirtualBox:~$
```

Monitoring disk usage helps prevent service interruptions

System logs were reviewed to monitor authentication attempts and system events. Logs provide an audit trail that can be used for troubleshooting and security analysis:

```
studentuser@katana-VirtualBox:~$ sudo journalctl -u ssh
[sudo] password for studentuser:
Dec 12 19:15:26 katana-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Dec 12 19:15:26 katana-VirtualBox sshd[5412]: Server listening on 0.0.0.0 port 22.
Dec 12 19:15:26 katana-VirtualBox sshd[5412]: Server listening on :: port 22.
Dec 12 19:15:26 katana-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Dec 13 13:32:24 katana-VirtualBox systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Dec 13 13:32:24 katana-VirtualBox sshd[5412]: Received signal 15; terminating.
Dec 13 13:32:24 katana-VirtualBox systemd[1]: ssh.service: Deactivated successfully.
Dec 13 13:32:24 katana-VirtualBox systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
Dec 13 13:32:24 katana-VirtualBox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Dec 13 13:32:24 katana-VirtualBox sshd[7694]: Server listening on 0.0.0.0 port 22.
Dec 13 13:32:24 katana-VirtualBox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Dec 13 13:32:24 katana-VirtualBox sshd[7694]: Server listening on :: port 22.
studentuser@katana-VirtualBox:~$
```

This showed SSH login attempts, including successful and failed connections. Log monitoring is essential for detecting suspicious activity.

To confirm service persistence, enabled services were listed:

```

studentuser@katana-VirtualBox:~$ systemctl list-unit-files --type=service
UNIT FILE                                STATE                                PRESET
accounts-daemon.service                 enabled                             enabled
alsa-restore.service                   static                             -
alsa-state.service                      static                             -
alsa-utils.service                     masked                             enabled
anacron.service                        enabled                             enabled
apparmor.service                       enabled                             enabled
apt-daily-upgrade.service               static                             -
apt-daily.service                      static                             -
autovt@.service                        alias                              -
avahi-daemon.service                   enabled                             enabled
blk-availability.service                enabled                             enabled
blueman-mechanism.service               enabled                             enabled
bluetooth.service                     disabled                             enabled
bolt.service                           static                             -
brltty-udev.service                   static                             -
brltty.service                         disabled                             enabled
casper-md5check.service                enabled                             enabled
casper.service                        enabled                             enabled
colord.service                         static                             -
configure-printer@.service              static                             -
console-getty.service                  disabled                             disabled
console-setup.service                  enabled                             enabled
container-getty@.service                static                             -
cron.service                           enabled                             enabled
cryptdisks-early.service                masked                             enabled
cryptdisks.service                     masked                             enabled
cups-browsed.service                   enabled                             enabled
cups.service                           enabled                             enabled
dbus-fi.wl.wpa_supplicant1.service      alias                              -
dbus-org.freedesktop.Avahi.service       alias                              -
dbus-org.freedesktop.hostname1.service  alias                              -
dbus-org.freedesktop.locale1.service    alias                              -
dbus-org.freedesktop.login1.service     alias                              -
dbus-org.freedesktop.ModemManager1.service alias                              -
dbus-org.freedesktop.nm-dispatcher.service alias                              -

```

This verified which services are enabled to start automatically at boot, ensuring system services behave as expected after a restart.

Week 6 successfully demonstrated the ability to manage system services, monitor system performance, and review logs for auditing purposes. These tasks are essential for maintaining a secure, reliable, and well-managed Linux system. Combined with previous weeks, this phase strengthens operational control and system oversight.